Variable Length Subnet Masking (VLSM)

Tree Method and Sequential Method

What is VLSM?

Variable Length Subnet Mask (VLSM) is an IP addressing technique that allows network administrators to divide an IP network into subnets of different sizes.

Key Concepts

1. **Traditional Subnetting (Fixed-Length)**
    - All subnets are the same size.
    - Often wastes IP addresses.
    - Example: Dividing `192.168.1.0/24` into 4 equal `/26` subnets (64 hosts each).
2. **VLSM (Variable-Length)**
    - Subnets can be different sizes.
    - Efficient use of IP address space.
    - Allocate close to what each network actually needs.
    - Example: One subnet with 100 hosts, another with 10 hosts.

Benefits of VLSM

- Reduces IP address waste.
- More efficient address allocation.
- Better scalability.
- Supports hierarchical network design.

Two Methods for VLSM Allocation

1. **Tree Method (Hierarchical Splitting)**
    - Start with the main network block.
    - Split into two equal halves repeatedly.
    - Allocate largest requirements first.
    - Visual, hierarchical approach.

2. **Sequential Method (Linear Allocation)**
    - Sort requirements from largest to smallest.
    - Allocate subnets one after another.
    - Calculate next available address after each allocation.
    - In this document, the sequential method reproduces the same final subnets as the tree method for each problem.

Important Formulas

- Usable hosts:
$$\text{Usable hosts} = 2^{(32-\text{prefix\_length})} - 2$$
- Network address: First address in the block.
- Broadcast address: Last address in the block.
- Usable range: Network + 1 to Broadcast - 1.

Practice Problems

This section contains three practice problems. For each problem you will see:

1. The problem description and requirement analysis.
2. The VLSM tree method solution (diagram) on a landscape page.
3. The sequential (linear) allocation solution that produces the *same subnets* as the tree method.

1 Problem 1 – Departmental Office Network

Scenario

A company has one office network that needs to be divided into departmental subnets on a single Class C private network. The requirements are:

- Sales Department – 60 hosts
- Marketing Department – 45 hosts
- IT Department – 10 hosts
- Guest Network – 5 hosts

Total: 120 hosts (plus some growth room).
Starting Network: `192.168.1.0/24`

Step 1: Analyze Requirements (Largest to Smallest)

Subnet Name	Hosts Needed	Required Prefix
Sales Dept	60	/25 (126 usable hosts)
Marketing Dept	45	/26 (62 usable hosts)
IT Dept	10	/28 (14 usable hosts)
Guest Network	5	/28 (14 usable hosts)

Solution 1A: Tree Method (Hierarchical Splitting)

```
192.168.1.0/24  (256 addresses)
|
+-- 192.168.1.0/25   (128 addresses) -> Sales Dept
|
+-- 192.168.1.128/25 (128 addresses) -> remaining
    |
    +-- 192.168.1.128/26 (64 addresses) -> Marketing Dept
    |
    +-- 192.168.1.192/26 (64 addresses) -> remaining
        |
        +-- 192.168.1.192/27 (32 addresses) -> remaining
        |   |
        |   +-- 192.168.1.192/28 (16 addresses) -> IT Dept
        |   +-- 192.168.1.208/28 (16 addresses) -> Available
        |
        +-- 192.168.1.224/27 (32 addresses) -> remaining
            |
            +-- 192.168.1.224/28 (16 addresses) -> Guest Network
            +-- 192.168.1.240/28 (16 addresses) -> Available
```

Allocated Subnets – Detailed Information

Subnet 1: Sales Department

- Hosts Required: 60
- Network Address: `192.168.1.0/25`
- Subnet Mask: `255.255.255.128`
- Broadcast Address: `192.168.1.127`
- Usable Host Range: `192.168.1.1` – `192.168.1.126`
- Usable Hosts: 126

Subnet 2: Marketing Department

- Hosts Required: 45
- Network Address: `192.168.1.128/26`
- Subnet Mask: `255.255.255.192`
- Broadcast Address: `192.168.1.191`

- Usable Host Range: `192.168.1.129` – `192.168.1.190`
- Usable Hosts: 62

Subnet 3: IT Department

- Hosts Required: 10
- Network Address: `192.168.1.192/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.1.207`
- Usable Host Range: `192.168.1.193` – `192.168.1.206`
- Usable Hosts: 14

Subnet 4: Guest Network

- Hosts Required: 5
- Network Address: `192.168.1.224/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.1.239`
- Usable Host Range: `192.168.1.225` – `192.168.1.238`
- Usable Hosts: 14

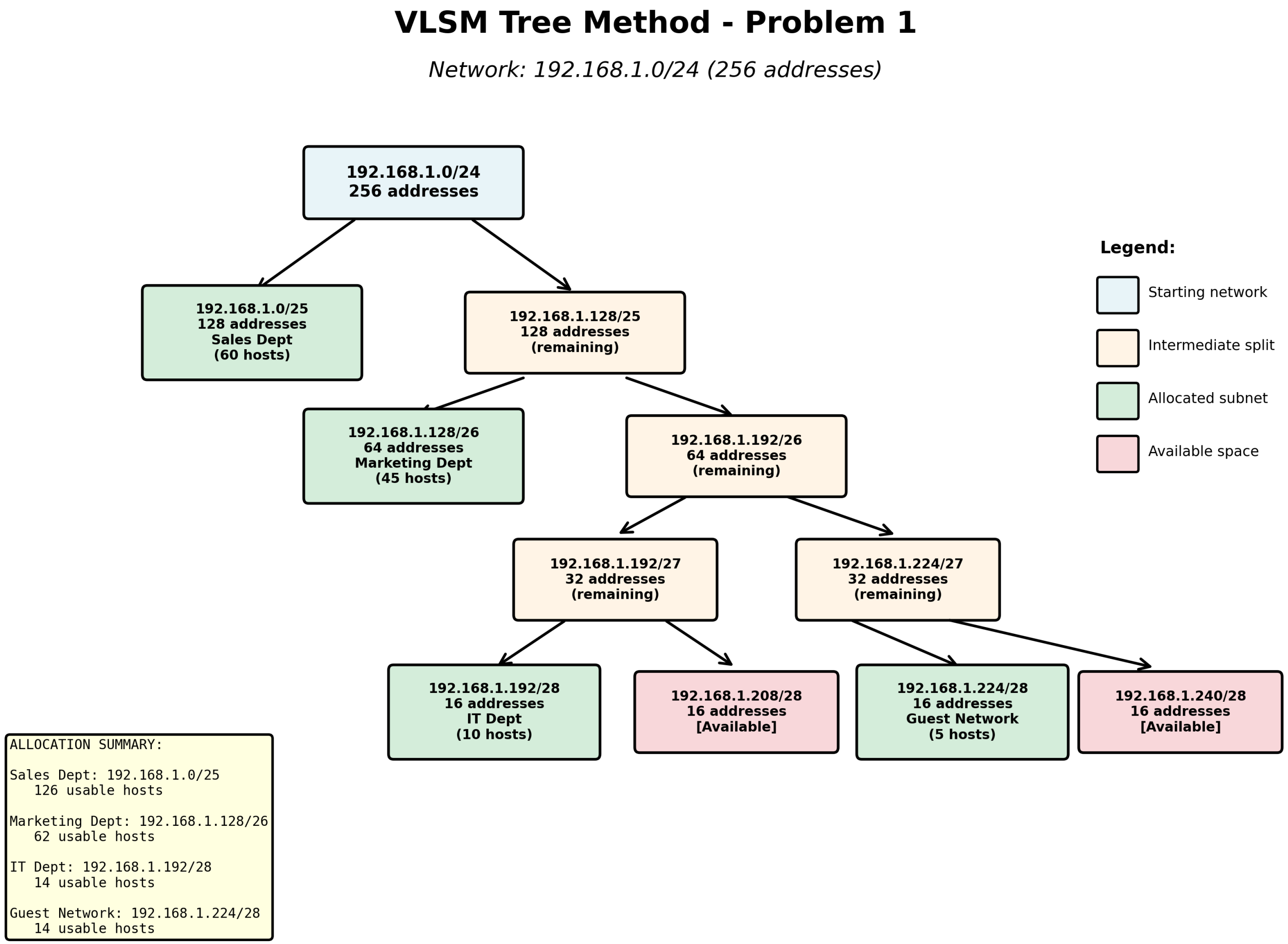

Figure 1: VLSM Tree Method – Problem 1 (Departmental Office Network)

Solution 1B: Sequential Method (Linear Allocation)

In this problem the sequential allocation is chosen so that the final subnets exactly match the tree diagram and the plot.

Allocation Process

Step 1: Allocate Sales Department

- Hosts Required: 60
- Prefix Needed: /25 (126 usable hosts)
- Starting Network: `192.168.1.0`
- Network Address: `192.168.1.0/25`
- Subnet Mask: `255.255.255.128`
- Broadcast Address: `192.168.1.127`
- Usable Host Range: `192.168.1.1` – `192.168.1.126`
- Block Size: 128 addresses
- Next Available IP: `192.168.1.128`

Step 2: Allocate Marketing Department

- Hosts Required: 45
- Prefix Needed: /26 (62 usable hosts)
- Network Address: `192.168.1.128/26`
- Subnet Mask: `255.255.255.192`
- Broadcast Address: `192.168.1.191`
- Usable Host Range: `192.168.1.129` – `192.168.1.190`
- Block Size: 64 addresses
- Next Available IP: `192.168.1.192`

Step 3: Allocate IT Department

- Hosts Required: 10
- Prefix Needed: /28 (14 usable hosts)
- Network Address: `192.168.1.192/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.1.207`
- Usable Host Range: `192.168.1.193` – `192.168.1.206`

- Block Size: 16 addresses
- Next Available IP: `192.168.1.208`

Step 4: Allocate Guest Network

- Hosts Required: 5
- Prefix Needed: /28 (14 usable hosts)
- Network Address: `192.168.1.224/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.1.239`
- Usable Host Range: `192.168.1.225` – `192.168.1.238`
- Block Size: 16 addresses

2 Problem 2 – Manufacturing Site Network

Scenario

A manufacturing site uses a single Class C private network and needs the following subnets:

- Production Floor – 100 hosts
- Office Staff – 30 hosts
- Server Room – 15 hosts
- Warehouse – 8 hosts

Total: 153 hosts
Starting Network: `192.168.2.0/24`

Step 1: Analyze Requirements (Largest to Smallest)

Subnet Name	Hosts Needed	Required Prefix
Production Floor	100	/25 (126 usable hosts)
Office Staff	30	/27 (30 usable hosts)
Server Room	15	/27 (30 usable hosts)
Warehouse	8	/28 (14 usable hosts)

Solution 2A: Tree Method (Hierarchical Splitting)

```
192.168.2.0/24  (256 addresses)
|
+-- 192.168.2.0/25   (128 addresses) -> Production Floor
|
+-- 192.168.2.128/25 (128 addresses) -> remaining
    |
    +-- 192.168.2.128/26 (64 addresses) -> remaining
    |   |
    |   +-- 192.168.2.128/27 (32 addresses) -> Office Staff
    |   +-- 192.168.2.160/27 (32 addresses) -> Server Room
    |
    +-- 192.168.2.192/26 (64 addresses) -> remaining
        |
        +-- 192.168.2.192/27 (32 addresses) -> to be split
        |   |
        |   +-- 192.168.2.192/28 (16 addresses) -> Warehouse
        |   +-- 192.168.2.208/28 (16 addresses) -> Available
        |
        +-- 192.168.2.224/27 (32 addresses) -> Available
```

Allocated Subnets – Detailed Information

Subnet 1: Production Floor

- Hosts Required: 100

- Network Address: `192.168.2.0/25`
- Subnet Mask: `255.255.255.128`
- Broadcast Address: `192.168.2.127`
- Usable Host Range: `192.168.2.1` – `192.168.2.126`
- Usable Hosts: 126

Subnet 2: Office Staff

- Hosts Required: 30
- Network Address: `192.168.2.128/27`
- Subnet Mask: `255.255.255.224`
- Broadcast Address: `192.168.2.159`
- Usable Host Range: `192.168.2.129` – `192.168.2.158`
- Usable Hosts: 30

Subnet 3: Server Room

- Hosts Required: 15
- Network Address: `192.168.2.160/27`
- Subnet Mask: `255.255.255.224`
- Broadcast Address: `192.168.2.191`
- Usable Host Range: `192.168.2.161` – `192.168.2.190`
- Usable Hosts: 30

Subnet 4: Warehouse

- Hosts Required: 8
- Network Address: `192.168.2.192/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.2.207`
- Usable Host Range: `192.168.2.193` – `192.168.2.206`
- Usable Hosts: 14

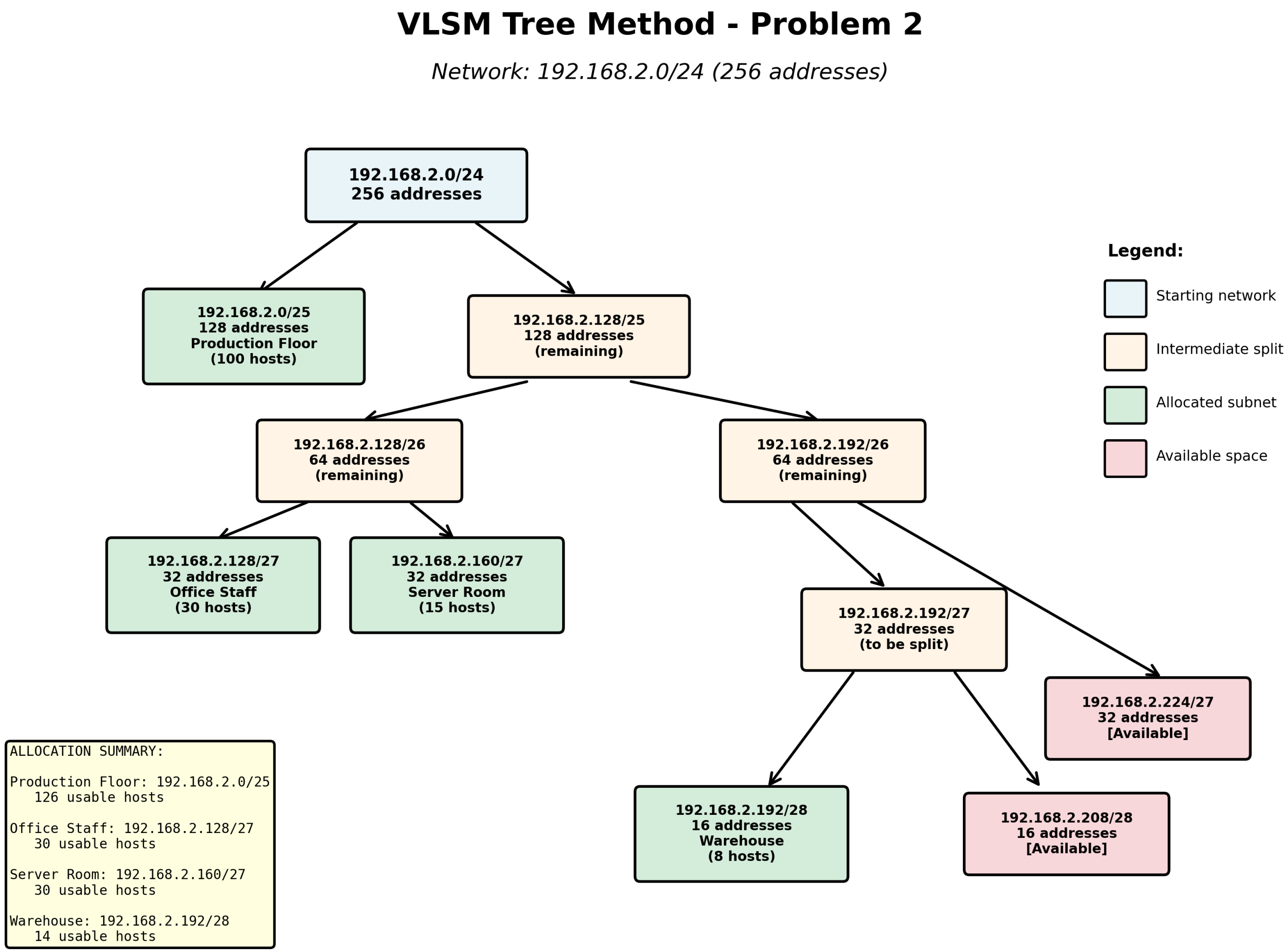

Figure 2: VLSM Tree Method – Problem 2 (Manufacturing Site Network)

Solution 2B: Sequential Method (Linear Allocation)

In this problem the sequential allocation follows the same blocks as the tree.

Allocation Process

Step 1: Allocate Production Floor

- Hosts Required: 100
- Prefix Needed: /25 (126 usable hosts)
- Network Address: `192.168.2.0/25`
- Subnet Mask: `255.255.255.128`
- Broadcast Address: `192.168.2.127`
- Usable Host Range: `192.168.2.1` – `192.168.2.126`

Step 2: Allocate Office Staff

- Hosts Required: 30
- Prefix Needed: /27 (30 usable hosts)
- Network Address: `192.168.2.128/27`
- Subnet Mask: `255.255.255.224`
- Broadcast Address: `192.168.2.159`
- Usable Host Range: `192.168.2.129` – `192.168.2.158`

Step 3: Allocate Server Room

- Hosts Required: 15
- Prefix Needed: /27 (30 usable hosts)
- Network Address: `192.168.2.160/27`
- Subnet Mask: `255.255.255.224`
- Broadcast Address: `192.168.2.191`
- Usable Host Range: `192.168.2.161` – `192.168.2.190`

Step 4: Allocate Warehouse

- Hosts Required: 8
- Prefix Needed: /28 (14 usable hosts)
- Network Address: `192.168.2.192/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.2.207`
- Usable Host Range: `192.168.2.193` – `192.168.2.206`

3 Problem 3 – Multi-Site Corporate Network (Complex)

Scenario

A company with multiple sites needs comprehensive network design:

- HQ LAN – 120 hosts
- Branch 1 LAN – 60 hosts
- Branch 2 LAN – 30 hosts
- Branch 3 LAN – 25 hosts
- Data Center – 50 hosts
- DMZ (Public Servers) – 10 hosts
- WAN 1 (HQ to Branch 1) – 2 hosts
- WAN 2 (HQ to Branch 2) – 2 hosts
- WAN 3 (HQ to Branch 3) – 2 hosts
- WAN 4 (HQ to Data Center) – 2 hosts

Total: 303 hosts
Starting Network: `192.168.100.0/23` (512 addresses available)

Step 1: Analyze Requirements (Largest to Smallest)

Subnet Name	Hosts Needed	Required Prefix
HQ LAN	120	/25 (126 usable hosts)
Branch 1 LAN	60	/26 (62 usable hosts)
Data Center	50	/26 (62 usable hosts)
Branch 2 LAN	30	/27 (30 usable hosts)
Branch 3 LAN	25	/27 (30 usable hosts)
DMZ (Public Servers)	10	/28 (14 usable hosts)
WAN 1 (HQ to B1)	2	/30 (2 usable hosts)
WAN 2 (HQ to B2)	2	/30 (2 usable hosts)
WAN 3 (HQ to B3)	2	/30 (2 usable hosts)
WAN 4 (HQ to DC)	2	/30 (2 usable hosts)

Solution 3A: Tree Method (Hierarchical Splitting)

(See detailed tree diagram on the next landscape page.)

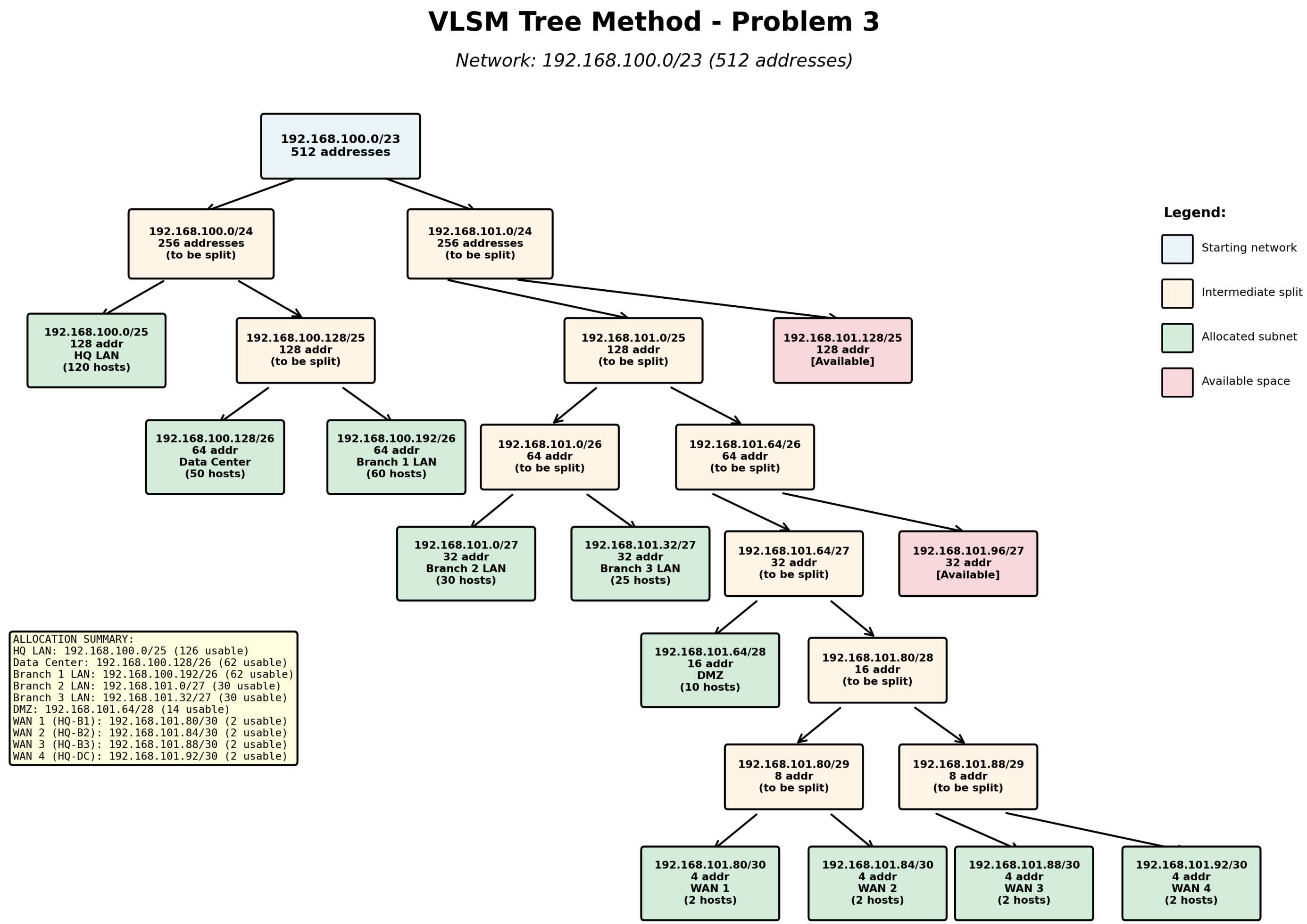

Figure 3: VLSM Tree Method – Problem 3 (Multi-Site Corporate Network)

Solution 3B: Sequential Method (Linear Allocation)

Here the sequential allocation is chosen so that the resulting subnets are identical to those in the tree diagram.

Allocation Process

Step 1: Allocate HQ LAN

- Hosts Required: 120
- Prefix Needed: /25 (126 usable hosts)
- Network Address: `192.168.100.0/25`
- Subnet Mask: `255.255.255.128`
- Broadcast Address: `192.168.100.127`
- Usable Host Range: `192.168.100.1` – `192.168.100.126`

Step 2: Allocate Data Center

- Hosts Required: 50
- Prefix Needed: /26 (62 usable hosts)
- Network Address: `192.168.100.128/26`
- Subnet Mask: `255.255.255.192`
- Broadcast Address: `192.168.100.191`
- Usable Host Range: `192.168.100.129` – `192.168.100.190`

Step 3: Allocate Branch 1 LAN

- Hosts Required: 60
- Prefix Needed: /26 (62 usable hosts)
- Network Address: `192.168.100.192/26`
- Subnet Mask: `255.255.255.192`
- Broadcast Address: `192.168.100.255`
- Usable Host Range: `192.168.100.193` – `192.168.100.254`

Step 4: Allocate Branch 2 LAN

- Hosts Required: 30
- Prefix Needed: /27 (30 usable hosts)
- Network Address: `192.168.101.0/27`
- Subnet Mask: `255.255.255.224`

- Broadcast Address: `192.168.101.31`
- Usable Host Range: `192.168.101.1` – `192.168.101.30`

Step 5: Allocate Branch 3 LAN

- Hosts Required: 25
- Prefix Needed: /27 (30 usable hosts)
- Network Address: `192.168.101.32/27`
- Subnet Mask: `255.255.255.224`
- Broadcast Address: `192.168.101.63`
- Usable Host Range: `192.168.101.33` – `192.168.101.62`

Step 6: Allocate DMZ (Public Servers)

- Hosts Required: 10
- Prefix Needed: /28 (14 usable hosts)
- Network Address: `192.168.101.64/28`
- Subnet Mask: `255.255.255.240`
- Broadcast Address: `192.168.101.79`
- Usable Host Range: `192.168.101.65` – `192.168.101.78`

Step 7: Allocate WAN 1 (HQ to Branch 1)

- Hosts Required: 2
- Prefix Needed: /30 (2 usable hosts)
- Network Address: `192.168.101.80/30`
- Subnet Mask: `255.255.255.252`
- Broadcast Address: `192.168.101.83`
- Usable Host Range: `192.168.101.81` – `192.168.101.82`

Step 8: Allocate WAN 2 (HQ to Branch 2)

- Hosts Required: 2
- Prefix Needed: /30 (2 usable hosts)
- Network Address: `192.168.101.84/30`
- Subnet Mask: `255.255.255.252`
- Broadcast Address: `192.168.101.87`
- Usable Host Range: `192.168.101.85` – `192.168.101.86`

Step 9: Allocate WAN 3 (HQ to Branch 3)

- Hosts Required: 2
- Prefix Needed: /30 (2 usable hosts)
- Network Address: `192.168.101.88/30`
- Subnet Mask: `255.255.255.252`
- Broadcast Address: `192.168.101.91`
- Usable Host Range: `192.168.101.89` – `192.168.101.90`

Step 10: Allocate WAN 4 (HQ to Data Center)

- Hosts Required: 2
- Prefix Needed: /30 (2 usable hosts)
- Network Address: `192.168.101.92/30`
- Subnet Mask: `255.255.255.252`
- Broadcast Address: `192.168.101.95`
- Usable Host Range: `192.168.101.93` – `192.168.101.94`